KAUNAS UNIVERSITY OF TECHNOLOGY INFORMATICS FACULTY

INTRODUCTION TO ARTIFICIAL INTELLIGENCE
LAB 2

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1. Work description

Artificial neural network with an elementary structure will be used during the work. The structure is a neuron with linear activation function (purelin(n) = purelin(Wp+b) = Wp+b). Neuron task is to forecast the k-th value of time series a(k) using n previous values a(k-1), a(k-2),...,a(k-n).

Model that is implemented using a premise that a dependency between a forecasted value and previous values can be described using a linear function is called autoregressive linear model of the n-th order.

2. Model Schema and Data

Linear autoregressive model has the following form: a_hat(k) = w1*a(k-1) + w2*a(k-2) + ... + wn*a(k-n) + b. Here w1, w2, wn and b are model parameters and a_hat(k) means the next forecasted value of time series. Forecasted value will be at the neuron output.

Sun plum activity on the sun will be forecasted in this work. The activity is expressed by a number of plums during a certain calendar year. This activity is cyclic and cycle is 11 years.

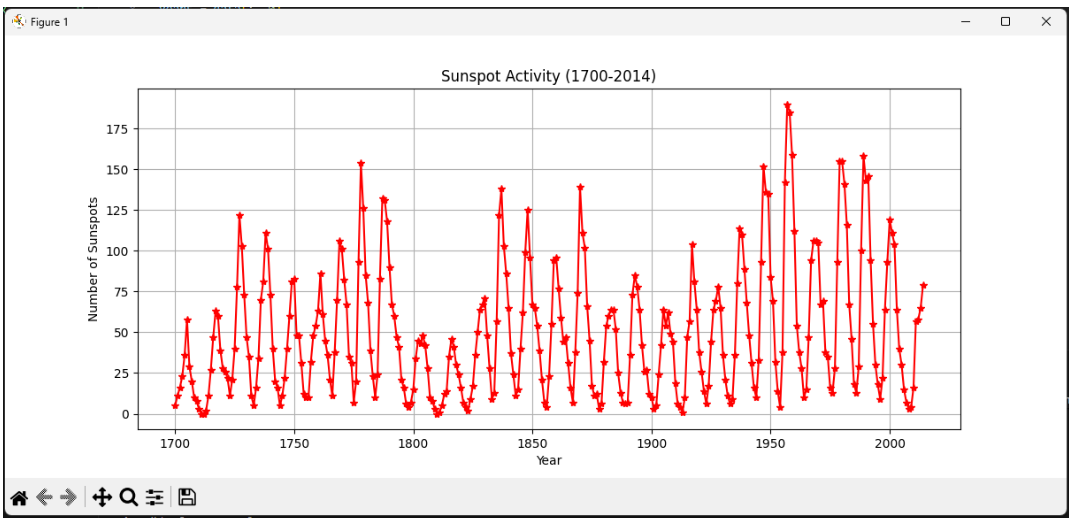

FIGURE 1.1: NEURON SCHEMA AND HISTORICAL DATA PLOT

3. Work Instructions and Experiments

Historical data on sun plum activity during 1700 - 2014 years was loaded into the working memory. The model order was initially set to n=2, meaning the forecast is based on two previous years.

The data set was split into training and test sets. The first 200 members were used for training to calculate optimal values of neuron weight coefficients. The weights were calculated using matrix calculus (newlind function) and iterative supervised learning (newlin and train functions).

4. Performance Metrics

Mean-Square-Error (MSE) and Median Absolute Deviation (MAD) were calculated to verify model forecast quality:

MSE = (1/N) * sum((a(k) - a_hat(k))^2)

MAD = median(|a(k) - a_hat(k)|)

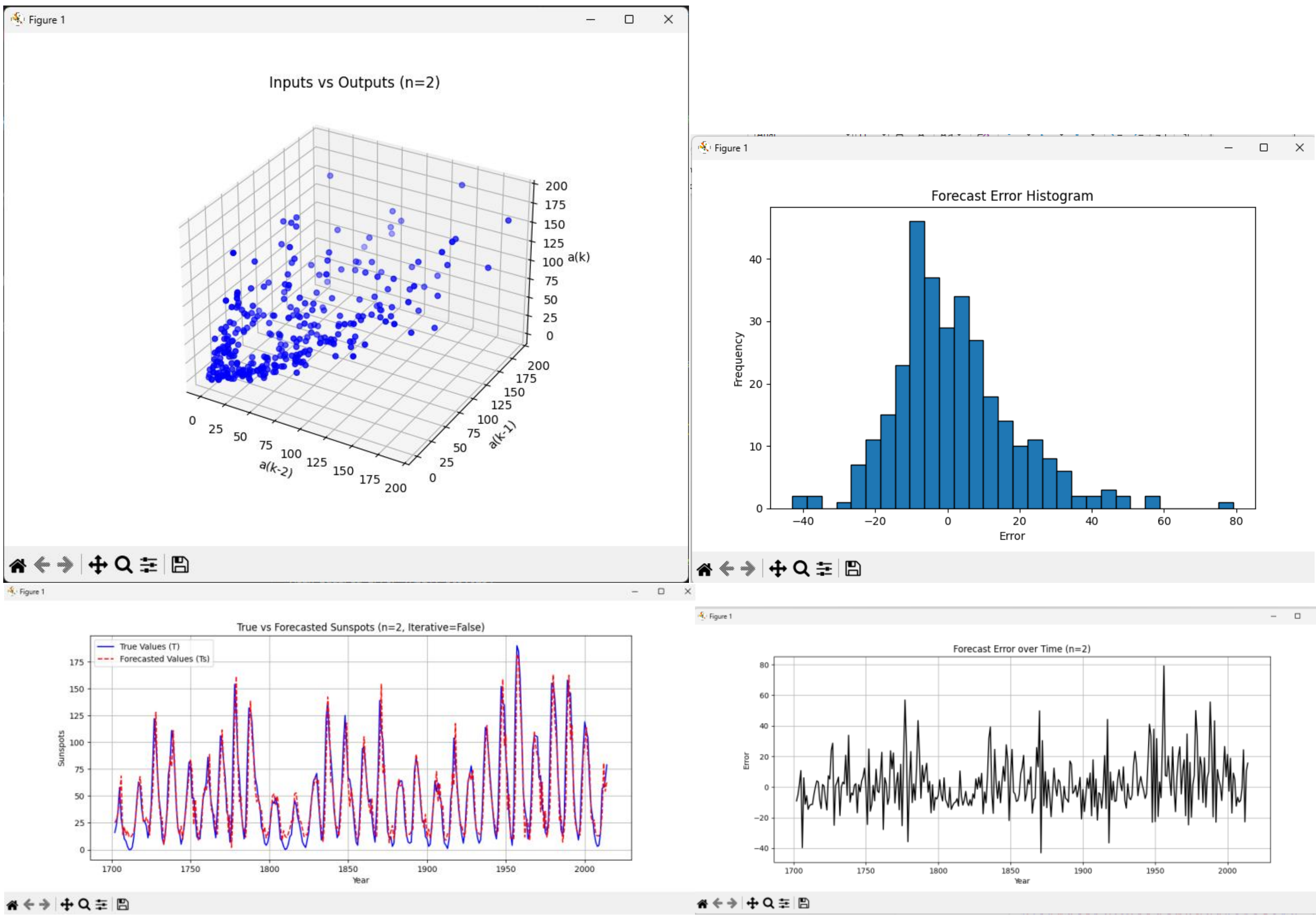

FIGURE 2: 3D INPUT/OUTPUT PLOT AND FORECAST COMPARISON

5. Model Structure Investigation

The number of network inputs was changed to n=6 and n=11 to investigate the impact of model structure change on forecast quality. Experiments with the newlin function were performed to analyze convergence and error rates across different input sizes.

6. Conclusions

The linear autoregressive model implemented by the artificial neuron effectively predicts the cyclic activity of sunspots. Matrix calculus provided an exact solution for optimal weights, while iterative training demonstrated the learning process through epochs. Increasing the model order (n=11) improved the alignment with the 11-year periodicity of the solar data. Accurate selection of the learning rate is essential for the iterative training process to converge without divergence.